

Wave spectrum estimation conditioned on machine learning-based output using the wave buoy analogy

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ABSTRACT

In this work, a hybrid approach for wave spectrum estimation is proposed. Fundamentally, the approach is based on the wave buoy analogy, processing ship response measurements, via a framework combining machine learning and a physics-based method dependent on available transfer functions. Specifically, a non-parametric (Bayesian) estimate is obtained of the directional wave spectrum conditioned on integral wave parameters established by a convolutional neural network. The developed method is assessed in a case study considering about two years of data obtained from an in-service container ship. The method produces good results, significantly improved when compared to the initial estimate made without constraints.

1. Introduction

Having an estimate of the on-site sea state is essential for reliable assessment of the performance of marine operations at sea; where both safety and energy efficiency are main concerns, addressed during in-service situations as well as in post-voyage analyses. As such, much effort has been given to developing methods and concepts from which the sea state can be estimated by processing measurements of wave-induced responses from shipboard sensors, in particular motions recorded by motion response units (MRUs). The literature refers to these methods and concepts under the name *the wave buoy analogy* or *ship as a wave buoy*; herein, the former is used [1]. As a practical remark, in this study, *sea state* generally refers to a characterisation of the wave system, and thus may refer to the complete directional wave spectrum or to integral wave parameters reflecting the statistics of the waves' height, period, and direction.

In most earlier applications, the wave buoy analogy has been formulated as a frequency-domain approach [e.g., 2–7], where it is understood that spectral analysis yields the relationship between measured response spectra, on the one side, and, on the other side, corresponding theoretically computed response spectra, obtained by the combination of (wave-to-response) transfer functions and the directional wave spectrum. Thus, the relationship establishes an equation system from which the unknown wave spectrum can be estimated, assuming the transfer functions to be available as per computations using potential theory codes or less sophisticated closed-form expressions if the detailed hull geometry is not at hand [8]. It can be said that this sort of frequency-domain approach is model-based, or physics-based, since pre-modelling of the ship motion dynamics, given in terms of transfer functions, is a fundamental requirement. However, in more recent studies, the wave buoy analogy has also been formulated in ways allowing the sea state to be estimated from data-driven, i.e. machine learning-based, approaches depending exclusively on the availability of extensive training data encompassing measurements of responses and corresponding sea state data, obtained e.g. via hindcasts or ship-mounted wave radars. In this way, machine learning circumvents the need for transfer functions.

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1.1. Objective and motivation

The fundamental approaches – the physics-based and the purely data-driven – of the wave buoy analogy have their own pros and cons, but, until now, no efforts have to the authors' knowledge attempted to blend the two approaches, in a hybrid framework. In a nutshell, the present study investigates a conceptual idea for doing exactly that by simply using a data-driven estimate of integral wave parameters as a constraint, or *condition*, on the wave spectrum estimate to be produced from a physics-based approach. The motivation for this merging is to obtain a more reliable estimate of the directional wave spectrum. Here, it should be noticed that machine learning-based frameworks [e.g., 9–13] have proven very capable in estimating integral wave parameters, whereas it is a more difficult and demanding task to obtain (directional) wave spectra, although research in this direction has been initiated [14]. It is left as a practical remark that, onwards, the term *condition* or *conditions* and related terms (conditioned, unconditioned, conditioning, etc.) are used consistently for referring to wave spectrum estimates by the wave buoy analogy, produced either with or without *conditions* obtained from an external source. Thus, the term(s) is used to refer to the fact that (ideally) the condition of the wave spectrum has been somehow improved.

1.2. Composition

Section 2 gives a concise introduction to the wave buoy analogy, and accounts are given of both the physics-based and machine learning-based frameworks, emphasising that the most basic aspects are included while many details are left to the specialised literature. Nevertheless, to understand the principle of the proposed hybrid method, the central theoretical aspects and the associated methodology are covered in Section 3. Section 4 introduces a case study considering an in-service container ship from which about two years of response data has been made available for assessing the proposed method. Section 5 contains the corresponding results and associated discussions, and Section 6 draws conclusions.

2. A condensed account of the wave buoy analogy

Fundamentally, the wave buoy analogy (from this point referred to by WBA) relies on measurements of wave-induced ship responses; say, heave (z), roll (ϕ), and pitch (θ). Introducing the linear time-invariant (LTI) assumption facilitates the use of transfer functions, which leads to what will be referred to as physics-based methods with formulations in both the frequency domain and the time domain. In contrast thereto, purely data-driven approaches exist, relying on machine learning, herein often referred to as ML-based methods. Common to all WBA methods is the characteristic of a ship as a wave filter, and the relative dimensions, notably the ratio of length to wavelength, indicate how well the measured responses can be used for wave inference.

As indicated above, the methods of the WBA normally consider *sets* of responses, but in the literature there is no agreed recommendation about exactly which responses to include, nor how many responses to include. From a theoretical point-of-view, however, it is clear that at least one response must be asymmetric with respect to waves approaching on port and starboard side, since it will otherwise not be possible, in the estimate of *wave direction*, to distinguish between waves approaching from the one side or the other side. Roll has this property, but so do other responses, and, in the case study presented later, sway (acceleration) is actually used, although the present section has mentioned roll. More general discussions about the inclusion and selection of responses and associated detailed analyses can be found in the literature [e.g., 1,3,4,7].

2.1. Physics-based methods

From a measured set of responses, corresponding response spectra $S_{i,j}(\omega_{e,l})$, $i, j = \{z, \phi, \theta\}$, can be computed under the assumption of stationary and ergodic processes. Here, $\omega_{e,l}$ defines the encounter frequency discretised by $l = 1, 2, \dots, L$ components. Formally, the directional wave spectrum $E(\omega, \mu)$ can be determined from the spectral equation,

$$\min \sum_{i,j} \sum_{l=1}^L \left| S_{i,j}(\omega_{e,l}) - \int_0^{2\pi} \left\langle \Phi_i(\omega, \beta) \overline{\Phi_j(\omega, \beta)} E(\omega, \mu) \frac{d\omega}{d\omega_e} \right\rangle_{\omega_{e,l}} d\mu \right|^2 \quad (1)$$

where the transfer function is $\Phi_i(\omega, \beta)$, and the overline denotes the complex conjugate. The intrinsic frequency is ω [rad/s], and $\beta = (180 + \chi - \mu)$ [deg] is the wave encounter angle, also referred to as the relative wave direction, with waves propagating from the compass direction μ while the compass heading of the ship is χ . The straight-line brackets $\langle \dots \rangle_{\omega_{e,l}}$, with $\omega_{e,l}$ as index, is used to emphasise that evaluation happens for a given frequency of encounter. Practical details related to an implementation of Eq. (1) are given in [15].

Frequency-domain solutions build on different assumptions and, typically, a distinction is made between non-parametric methods and parametric methods. Non-parametric methods solve for the unknown wave spectrum $E(\omega, \mu)$ at all discrete pairs of (ω, μ) , necessitating regularisation techniques and/or iterative schemes for dealing with the highly under-determined equation system. In the parametric methods, the directional wave spectrum is formed by an idealised wave spectrum, e.g., Bretschneider, JONSWAP or summations of such, overlaid with a spreading function. Thus, in the parametric methods, the minimisation of Eq. (1) leads to an optimised set of wave parameters governing the idealised wave spectrum. In the spectral equation, it is an inherent complexity that the Doppler shift, for ships with forward speed, introduces a 1-to-3 relationship between the encounter frequency and the intrinsic frequency for all wave encounter angles between following waves and beam waves. Although mathematically elementary, the Doppler shift is a practical complication and care must be taken when it is implemented in the WBA. As indicated by Eq. (1), several responses are used simultaneously, and the cross spectra reveal the necessary information about the phase relationships needed to infer the (relative) direction of the waves.

2.2. Machine learning-based methods

Modelling the ship motion dynamics via the governing physics necessitates several assumptions that can be relaxed if the relation between waves and motions instead is explored and determined entirely through data. In ML-based methods, the general relationship between waves and induced responses of a particular ship is learned by comprehensive training with large datasets of measured responses against available sea state information. The main advantage of ML-based methods is that transfer functions are not needed, and all associated uncertainties are thus removed. On the other hand, the need for high-quality sensor measurements is emphasised and so is the importance of having accurate (“external”) sea state information at the exact spatio-temporal position of the ship. Generally, sea state information can be obtained, for instance, through dedicated wave radar systems or via third-generation spectral wave models, such as WaveWatch III or WAM, often assimilated with data in re-analyses.

Many ML architectures allow, with relatively little effort through supervised regression, estimation of integral wave parameters, e.g., the significant wave height, the zero-upcrossing period, and the mean relative wave direction. As with the physics-based approaches, ML methods can be formulated in both the time domain and in the frequency domain; stressing the importance in having the phase relationships between responses, like with the physics-based methods, for obtaining reliable estimates of the (relative) wave direction.

2.3. Motivation of study revisited

Summarising some of the pros and cons of the two different frameworks (physics-based and ML-based) may serve to revisit the motivation for the study. As such, the following characteristics are noteworthy: (1) ML methods offer reliable estimates of integral parameters and there is no need for the transfer functions. (2) Physics-based methods can efficiently be applied for estimation of directional wave spectra. For instance, the samples presented in Fig. 1 are the results of the non-parametric method [4,6] applied to response measurements of a 2,800 TEU container ship, as documented in Section 4. In the figure, each subplot includes the directional wave spectrum estimates by WBA and Wavex, respectively, for specific time stamps. In fact, the WBA estimates are obtained by the herein proposed hybrid framework. (3) Transfer functions reflect the governing physics, and physics-based methods have the capability to reduce negative effects associated to scarcity in data of particular observations not included in the (training) dataset of ML methods. (4) The literature [e.g., 16–19] contains evidence that the physics-based methods may not always produce good estimates of the *directional* distribution of the wave energy density; that is, the (relative) wave direction is the most difficult to estimate correctly. In contrast, ML-based methods prove fairly reliable in the estimate of the relative wave direction.

Having these characteristics in mind, it appears interesting to form a hybrid model with the potential to emphasise the pros and minimise the cons of the two respective frameworks when they are merged in the proposed “ML-informed physics-based method”, reported about in the next section.

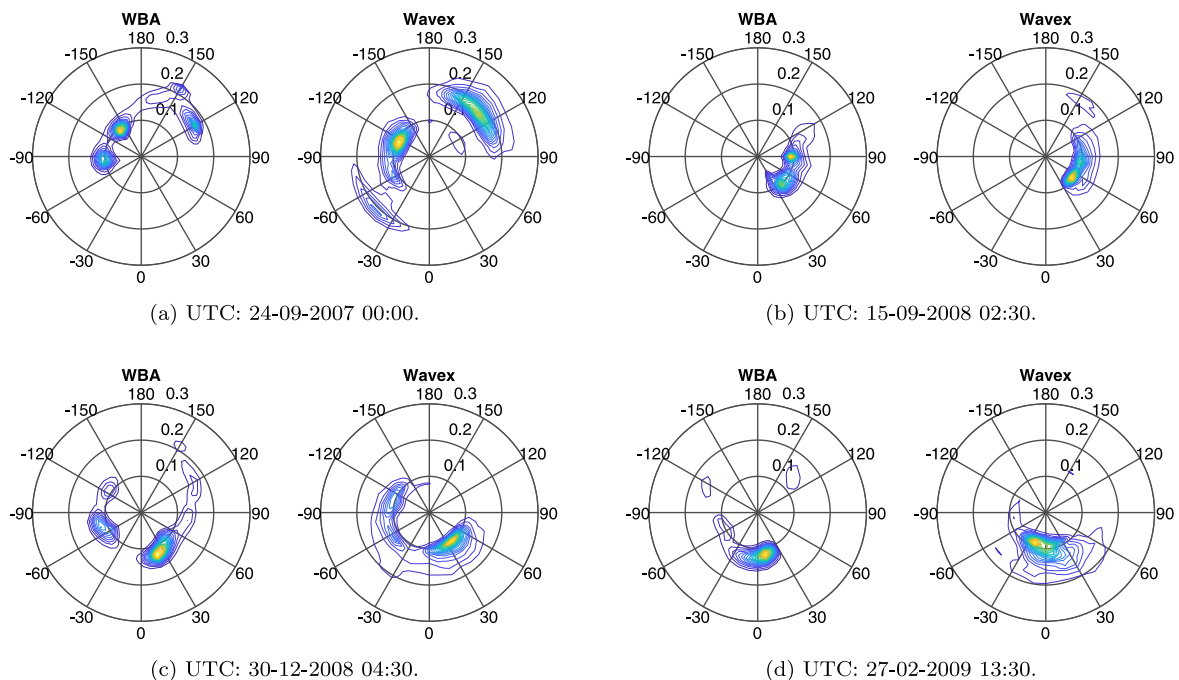


Fig. 1. Examples of directional wave spectra by the wave buoy analogy (WBA) and corresponding spectra by Wavex for specific time stamps (UTC). The WBA spectra are conditioned on outcome of machine learning, see Section 3. Note that the polar plots present where the waves *come from*, relative to the ship, with 180° being head sea.

3. Theoretical aspects and methodology

3.1. Integral wave parameters

The directional wave spectrum $E(\omega, \mu)$ yields a complete description of how the energy density of a given (ocean) wave system is distributed as function of frequency and direction. Integration of the directional spectrum leads to more condensed representations, and the following expressions are all standard notations [e.g., 20], listed here for the sake of convenience. It should be noted that below list by no means is exhaustive. Furthermore, in the remainder of the paper, the terms *sea state parameters* and *integral wave parameters* are used interchangeably.

The (one-dimensional) frequency wave spectrum is obtained by integrating the directional wave spectrum with respect to direction:

$$F(\omega) = \int E(\omega, \mu) d\mu \quad (2)$$

and the spectral moment of order n is defined by

$$m_n = \int \omega^n F(\omega) d\omega \quad (3)$$

In practical computations, integration limits are needed although they are omitted in the formulas. Thus, it is understood that any integration over direction μ is made over the full 360-degrees range, i.e. from 0 to 2π . In principle, the frequency ω spans from 0 to infinity but, in practice, it is necessary to work with both a lower and an upper cut-off frequency. Additional remarks about this, applicable to the specific analysis, are given later.

The significant wave height H_s and the zero-upcrossing period T_z are given as:

$$H_s = 4\sqrt{m_0} \quad (4)$$

$$T_z = 2\pi\sqrt{\frac{m_0}{m_2}} \quad (5)$$

In addition, the peak period is

$$T_p = \frac{2\pi}{\omega_p}, \quad F(\omega_p) = \max_{\omega} F(\omega) \quad (6)$$

The mean relative wave direction follows from

$$D_m = \arctan(d/c) \quad (7)$$

where d and c are defined according to

$$\begin{aligned} d &= \iint E(\omega, \mu) \sin \mu d\omega d\mu \\ c &= \iint E(\omega, \mu) \cos \mu d\omega d\mu \end{aligned} \quad (8)$$

3.2. Identification of sea state parameters using deep learning

In the literature, several machine learning methods have been proposed for the estimation of sea state parameters. The present work relies on the method developed by Mittendorf et al. [13], in which a deep learning framework is proposed using sequential data in terms of response *spectra*. Thus, the inverse mapping, considering several ship responses, to prevalent sea conditions, as governed by significant wave height (H_s), peak period (T_p), and mean relative wave direction (D_m), is achieved by deep neural networks. Specifically, [13] showed that good results can be obtained through a convolutional neural network formulated in the frequency-domain and applied in a multi-task learning setting, where training data comprised corresponding datasets of measured 6 degrees-of-freedom motion responses and X-band (wave) radar data. In fact, the particular study [13] investigated the exact same case ship and data, as will be studied later in the paper, refer to Section 4. At this point, however, it suffices to note that sea state parameters, can be reliably estimated via the deep learning framework developed by Mittendorf et al. [13]. For the interested reader, additional details about the convolutional neural network are included in Appendix B, where it is noticed that the network relies on the so-called Inception architecture.

3.3. Wave spectrum estimation with conditions

The present work considers the non-parametric (“Bayesian”) approach by Nielsen [4,6] as the fundamental mathematical method from which to estimate directional wave spectra.¹ The original work [21] describes the derivation of the governing equation system

¹ It is noteworthy that in the remaining parts of the paper, the term *Bayesian* method is used as reference, if relevant, for pointing to the mathematical model of the WBA.

in details and also suggests how to impose conditions on the directional wave spectrum estimate by introducing additional equations expressing energy conservation between the measured and estimated *responses*. A short summary, presenting an outline of the Bayesian approach, is included in [Appendix A](#) for completeness. In addition, it appears appropriate to repeat and emphasise the inherent strength of this particular estimation concept for introducing how to condition the directional wave spectrum estimate on outcomes, i.e. *wave estimates*, from another source, like, for example, a machine learning model.

The great advantage of the Bayesian estimation concept [22,23] is the possibility to include *prior information* used for conditioning the solution of the governing equations. In qualitative terms, this allows to set conditions, for instance, on the second order derivative, controlling smoothness of the (directional) wave spectrum. Mathematically, in the Bayesian method, the inclusion of conditions leads to consideration of additional equations next to the governing (“physical”) equation expressed by Eq. (1), repeated here in matrix form:

$$\min \chi^2(\mathbf{x}) \equiv \min \|\mathbf{A}\mathbf{f}(\mathbf{x}) - \mathbf{b}\|^2 \quad (9)$$

where $\|\cdot\|$ represents the L_2 norm, and with the notation following from [Appendix A](#), stressing that the vector function $\mathbf{f}(\mathbf{x})$ expresses the unknown values of the wave spectrum $E(\omega, \mu)$ through a non-negativity constraint $\mathbf{f}(\mathbf{x}) = \exp(\mathbf{x})$.

In the summary, cf. [Appendix A](#), the inclusion of the additional set of equations, in terms of *priors* regularising the smoothing of the wave spectrum, is established via the functionals:

$$\varepsilon_{1mn}^2 = \mathbf{x}^T \mathbf{H}_1 \mathbf{x} \quad (10)$$

$$\varepsilon_{2mn}^2 = \mathbf{x}^T \mathbf{H}_2 \mathbf{x} \quad (11)$$

where the $\mathbf{H}$ -matrices relate to the sum of the second order difference of the unknown vector $\mathbf{x}$. Note that *two* functionals are used to regularise smoothing in two dimensions (frequency ω as well as direction μ). Moreover, the symbols on the left-hand sides should be understood as *concepts*, without mathematical definitions of the lower and upper indices.

In principle, the optimum solution, i.e. the wave spectrum estimate, is obtained by solving the constrained least squares problem [24] given by the minimisation of $S(\mathbf{x})$ with:

$$S(\mathbf{x}) \equiv \|\mathbf{A}\mathbf{f}(\mathbf{x}) - \mathbf{b}\|^2 + \mathbf{x}^T (u^2 \mathbf{H}_1 + v^2 \mathbf{H}_2) \mathbf{x} \quad (12)$$

In practice, however, optimum values of the hyperparameters u and v need be chosen, and this is basically achieved by simultaneously minimising a control criterion, as summarised in [Appendix A](#) and referring to [4,6] and, particularly, [21] for details.

Following an idea proposed in the original work [21], the present work suggests to condition the wave spectrum estimate by formulating conditions related directly to the distribution of *wave energy density* of the actual wave spectrum; not just the energy associated with the *responses*. In theory, conditions can be imposed corresponding to the number of integral wave parameters produced by the machine learning model. However, in this study, conditions are introduced via, respectively, the significant wave height and the mean relative wave direction, and the conditioning of the wave spectrum is thus made through the following equations:

$$\iint \tilde{E}(\omega, \mu) d\omega d\mu = \frac{1}{16} \hat{H}_s^2 \quad (13a)$$

$$\iint \tilde{E}(\omega, \mu) \sin \mu d\omega d\mu = \hat{d} \quad (13b)$$

$$\iint \tilde{E}(\omega, \mu) \cos \mu d\omega d\mu = \hat{c} \quad (13c)$$

where Eqs. (13b) and (13c) in combination relate to the mean relative wave direction, cf. Eq. (7). It is understood that the wide-tilde, in $\tilde{E}(\omega, \mu)$, is used to indicate the directional wave spectrum to be estimated, whereas the hat-notation on the right-hand side of the equations is used to indicate that the parameters are known from another source; in this case, they are the output from the machine learning model, i.e. the convolutional neural network.

Essentially, the conditioning through Eqs. (13a)–(13c) could be considered as additional priors, associating functionals and hyperparameters on their own. Yet, a different strategy is followed herein, and Eqs. (13a)–(13c) are considered as physical conditions which means that the equations are concatenated to the governing equation system represented by Eq. (9), which itself is the matrix variant of Eq. (1). In practice, this means that the coefficient matrix $\mathbf{A}$ is extended by three rows where the single elements, formally, are established from the coefficients governing the double-integration on the left-hand sides of Eqs. (13a)–(13c). Similarly, the vector $\mathbf{b}$ has three additional elements concatenated, reflecting the right-hand sides of Eqs. (13a)–(13c). Finally, it should be noted that the three new (conditioning) equations are weighted by a factor of 10 compared to the other equations in the governing equation system, cf. Eq. (9); with inspiration to the original work [4] where equivalence of *response* energy was introduced. The selection of the particular value (10) of the weight was made from a “subjective” trial and error analysis, with only a few different values (1, 10, 50) for a few different samples. As such, it has not been investigated if the value of the weight might be data and/or ship-dependent.

3.4. Assessment of estimates

Fundamentally, two outcomes of the WBA will be assessed: (1) Without conditioning, corresponding to the “raw” wave spectrum estimate by the Bayesian approach; and (2) With conditioning. It is interesting to include (1) for assessing the level of improvement by introducing the conditioning (2). With respect to (2), first, the actual conditioning is made using directly the results of the Wavex wave radar, see Section 4; this is to obtain a sort of validation, not affected by uncertainties involved when considering the output from the convolutional neural network. Ultimately, however, *test data* from the machine learning model [13] is used to condition the wave spectrum estimate of the Bayesian method.

In general, the involved uncertainties can be significant when estimating waves. To limit and study the effects of any bias or other types of uncertainties, estimates from an external source are included in the assessment. Thus, ERA5 estimates [25] act as a benchmark against the estimates produced by the developed hybrid model. In addition, it is also of interest to compare how well the wave spectrum estimates from different sources, e.g. Wavex vs. WBA, can be used for the prediction of a ship response *not* being part of the input for the wave estimation process itself. This type of comparison is included by considering predictions of the vertical (wave-induced) bending moment amidship and, subsequently, comparing to actual measurements. The theoretical prediction of a response spectrum follows from, referring to [15] for an actual algorithm carrying out the numerical implementation:

$$\hat{S}_R(\omega_e) = \int_0^{2\pi} |\Phi_R(\omega_e, \beta)|^2 E_e(\omega_e, \mu) d\mu \quad (14)$$

$$\beta = 180 - (\chi - \mu) \quad (15)$$

$$\omega_e = \omega_0 - \omega_0^2 \psi, \quad \psi = \frac{U}{g} \cos \beta \quad (16)$$

$$E_e(\omega_e, \mu) = \left\langle E(\omega_0, \mu) \frac{d\omega_0}{d\omega_e} \right\rangle_{\omega_e} \quad (17)$$

$$\Phi_R(\omega_e, \beta) = \langle \Phi_R(\omega_0, \beta) \rangle_{\omega_e} \quad (18)$$

where the syntax and notation is as with Eq. (1). From the spectrum, the 0th order moment follows immediately from

$$m_0 = \int \hat{S}_R(\omega_e) d\omega_e \quad (19)$$

The comparisons made later of the obtained integral wave parameters are to a wide extent made using correlation-type of plots, so that a line of identity means perfect agreement (if it were to be). To assess quantitatively the deviation between corresponding pairs of estimates, root-mean-squared-errors (RMSE), in dimensional form, are also computed for $\{H_s, T_p, T_z\}$, while the minimum-absolute-error² (MAE) is computed for D_m , accounting for the directional ambiguity. The expressions follow from:

$$\text{RMSE} = \sqrt{\frac{1}{K} \sum_{k=1}^K (y_{\alpha,k} - y_{\beta,k})^2} \quad (20)$$

$$\text{MAE} = \frac{1}{K} \sum_{k=1}^K \min\{\varepsilon_k, (360^\circ - \varepsilon_k)\} \quad (21a)$$

$$\varepsilon_k = |D_{m,\alpha,k} - D_{m,\beta,k}| \quad (21b)$$

where y_k indicates an estimate of the particular parameter in question, i.e. $\{H_s, T_p, T_z\}$ for sample k , using the methods α and β , representing pairs of methods: Wavex vs. ERA5, WBA vs. Wavex, or WBA vs. ERA5.

4. Case study

4.1. Vessel and route profile

The study builds on data from a 2800 TEU Panamax container vessel built in 1998. The vessel is pictured in Fig. 2, and its main particulars are listed in Table 1. During a four-year period between August 2007 and July 2011, the vessel was equipped with an extensive sensor framework and sailed in cross-Atlantic service. The data acquisition period was conducted over 4 years, but the installed wave radar (Wavex by Miros) was only in service until March 2009. Hence, the span of the data used herein is a little less than two years.

In Fig. 3, the GPS history of the case ship is visible and the effects of seamanship and weather routing [27,28] stand out since several routes deviate significantly from the shortest route (great circle) in the ship's trade between Western Europe and Quebec in Canada. The ship obviously operates in coastal and restricted waters, such as the St. Lawrence river; however, the focus of the present study is on deep-water conditions exclusively. For this reason, samples possibly influenced by shallow water and coastal effects are disregarded by enforcing spatial boundaries at -55° and -5° longitude as indicated in Fig. 3.

² Note the somewhat unconventional meaning of the abbreviation.



Fig. 2. Photograph of the considered vessel. From [26].

Table 1

Main particulars of the 2800 TEU container vessel.

Characteristic	Symbol	Unit	Value
Length between particulars	L_{pp}	[m]	232.0
Breadth	B	[m]	32.2
Design draught	T	[m]	10.78
Block coefficient	C_B	[-]	0.685
Deadweight	dwt	[t]	40,900
Brake power	P_B (@ 93 rpm)	[kW]	25,786

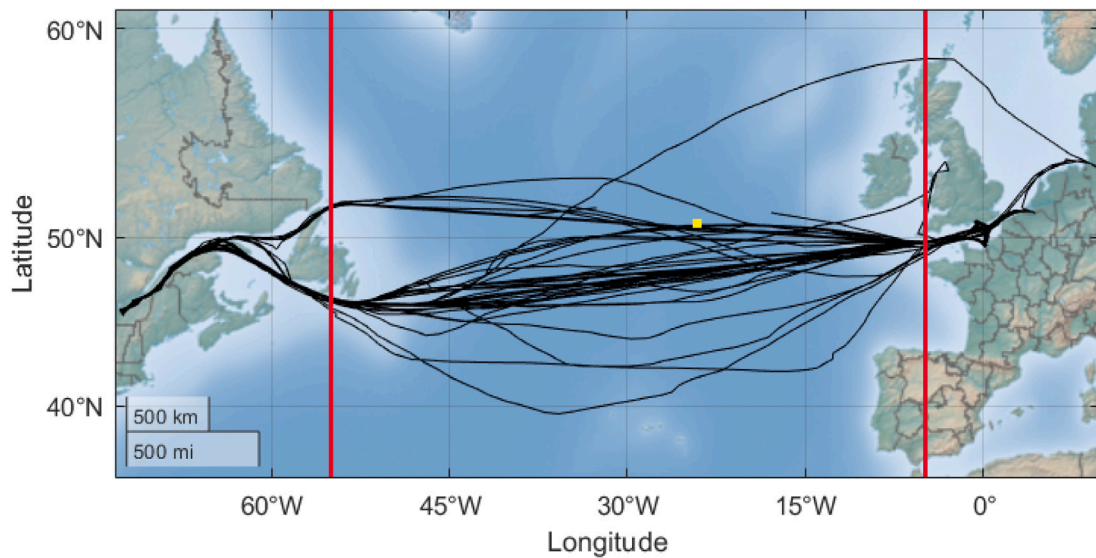


Fig. 3. GPS history (black) and the applied spatial boundaries (red). The yellow dot indicates one particular measurement sample. From [13].

4.2. Sensors and data collection

The case ship's on-board measurement system comprises several sensors, and the basic characteristics of those of importance to this study are briefly outlined in the following.

The Miros Wavex directional wave radar provides an averaged spectrum over a 25-min interval based on 1-min sample periods. Generally, X-band marine radars offer a relatively high spatio-temporal resolution. However, the accurate estimation

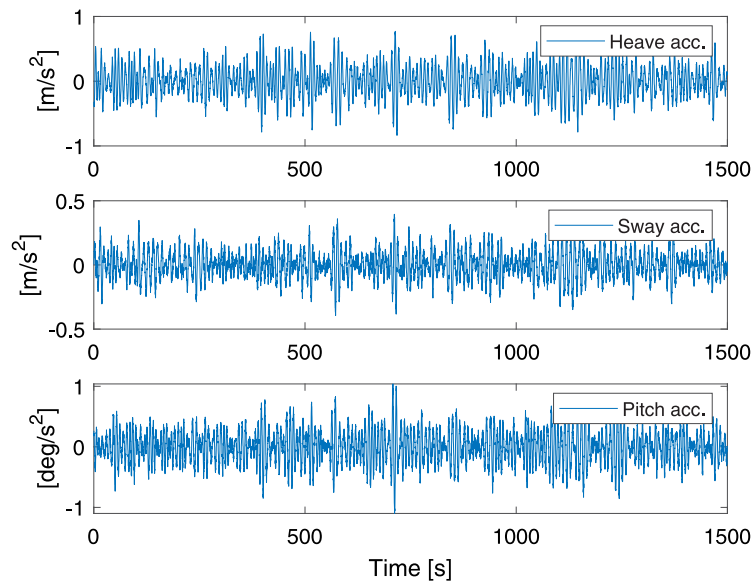


Fig. 4. Measured accelerations corresponding to the time indicated by the yellow dot in Fig. 3.

of the significant wave height is a matter of empirical corrections [29] and also the effect of rain clutter on data quality is noteworthy [30]. The Motion Response Unit (MRU), measuring 6 DOF motions (i.e. accelerations of surge, sway, heave, roll, pitch, and yaw, respectively), is located 78.5 m forward of the aft perpendicular (AP), 11.7 m above base line (BL), and at the centre line (CL). The MRU was installed in a socket, filtering high-frequency noise in the range of 50–100 Hz caused by e.g. thrusters or pumps. For wave estimation using the WBA, time series of 25 min duration are considered, and this study makes estimates using the simultaneous combination of translational and rotational accelerations corresponding to {heave, sway, pitch}. An example of a 25-min period of the recorded motion time series is presented in Fig. 4. To construct the cross spectra, needed as input to the WBA, FFT is performed using the Matlab function `cpsd`, which is based on Welch's averaged, modified periodogram method applied on eight sections with 50% overlap, and with each section windowed with a Hamming window [31].

In addition to the MRU, four strain gauges were installed on the longitudinal stiffeners at port and starboard in the aft section (50.3 m from AP) and amidships (118.7 m from AP). As described by Storhaug et al. [26], the port and starboard strain sensors are aggregated into two artificial or virtual sensors indicating pure vertical bending and axial stress in the aft and midship sections. Lastly, as already introduced, the GPS position as well as the Speed Over Ground (SOG) were obtained during the data acquisition period.

It is noteworthy that draught and trim were not measured. Storhaug et al. [26] state that the typical transit draught is approximately 9.5 m and the transit trim lies between 0.5 m to 1.0 m. In addition, the loading condition, and thus the transverse metacentric height GM_T , is not included in the dataset, adding further uncertainty due to the widespread operational profile of container vessels, in general.

4.3. Transfer functions

The transfer functions of ship motions as well as structural loads have been computed by DNV's software for linear and non-linear hydrodynamic analysis [32], which solves the fully 3-dimensional radiation-diffraction problem by a Rankine panel method. As pointed out at the end of the previous subsection, the variation in draught, occurring over the considered voyages, is unknown, and the transfer functions are computed for one draft only, herein taken as the typical transit draught (9.5 m). On the other hand, assessing the speed profile, see Fig. 5, reveals that the vast majority of data has been obtained within the speed range 17–21 knots. Consequently, it has been decided to compute transfer functions at discrete speeds $U_s = \{17, 18, 19, 20, 21\}$ knots, selecting the nearest speed in the actual estimation analyses. In doing so, it has to be acknowledged that deviations larger than 0.5 knots between the measured speed and the 'transfer function speed' is possible when the measured speed is lower than 17 knots, respectively, higher than 21 knots. In this context, it is noticed that the *measured* speed is always taken as the average over the considered 25 min time series.

Examples of the transfer function amplitudes for heave, sway, and pitch accelerations are shown in Fig. 6, noticing that the amplitudes reflect the speed $U_s = 17$ knots. Any assessment of the transfer functions is not attempted herein. The central point to note is that, in reality, the transfer functions should have been computed for a vast number of different operational conditions to reflect the ship's actual operational profile. The study's limitation in this direction obviously introduce a significant uncertainty, and it appears relevant, as a future work, to conduct dedicated studies about the associated effects on outcomes of the wave spectrum estimates.

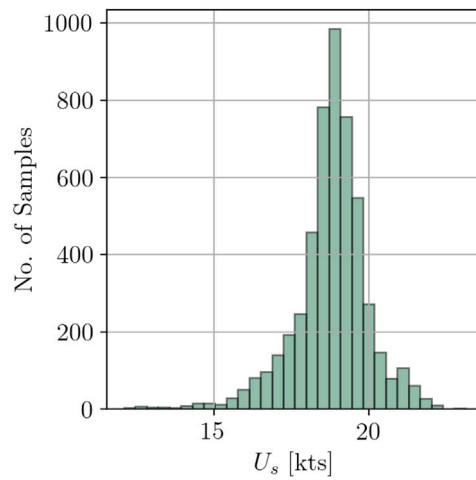


Fig. 5. Histogram of the ship advance speed U_s according to GPS data.

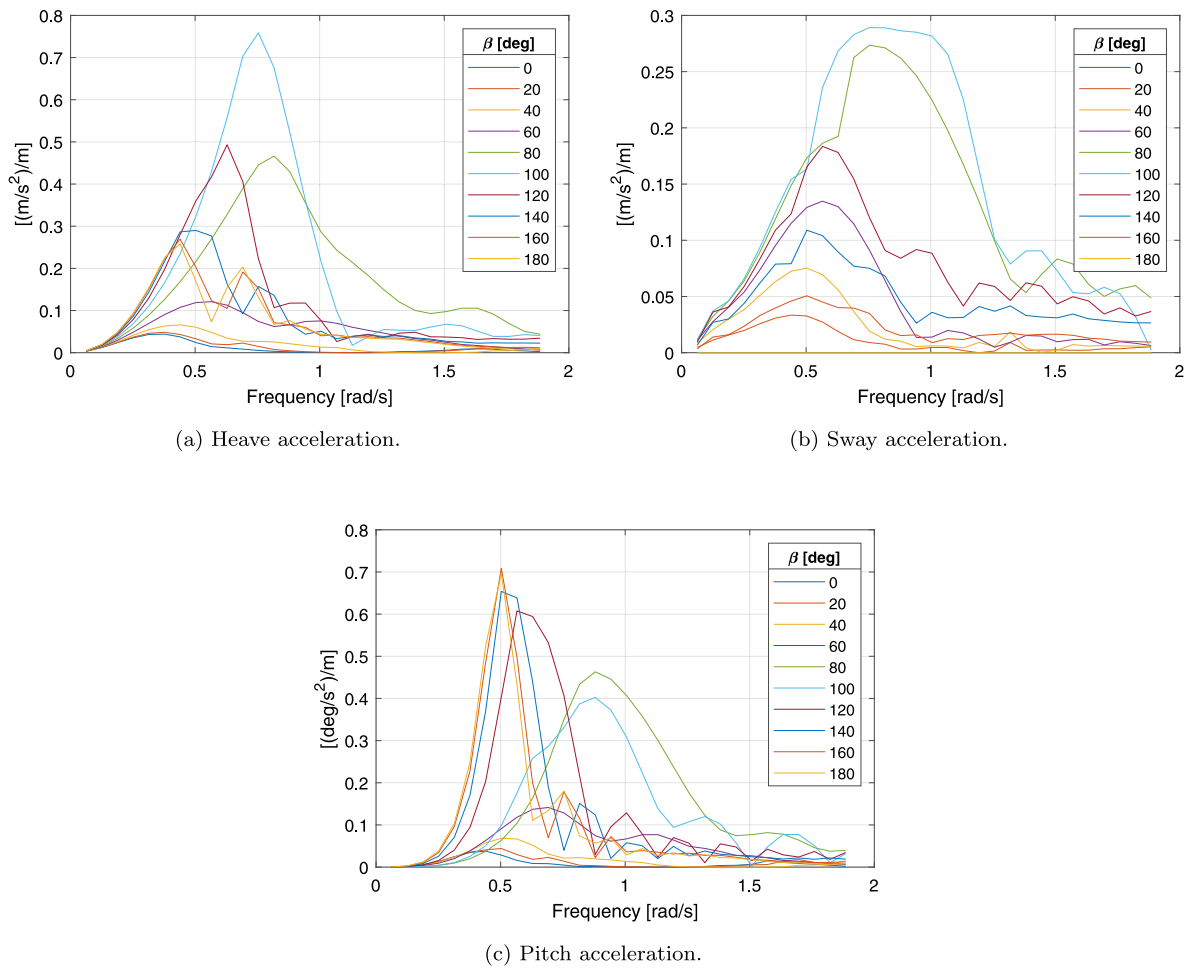


Fig. 6. Amplitudes of the transfer functions used for wave spectrum estimation. The amplitudes correspond to a speed $U_s = 17$ knots. The effect of variations in the wave encounter angle (β) is included.

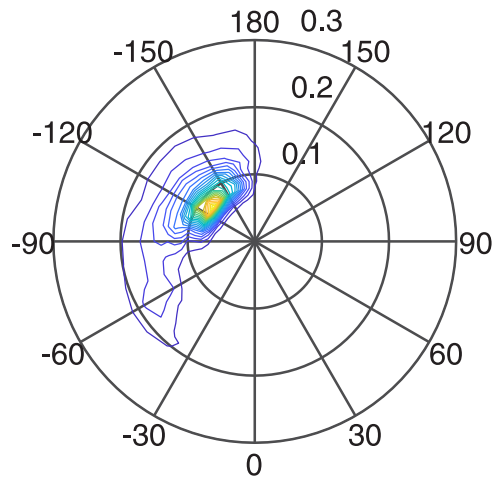


Fig. 7. Wavex directional wave spectrum corresponding to the time series in Fig. 4.

4.4. Sea state estimates from marine X-band radar (Wavex)

Generally, it has little meaning to speak about a ground truth, considering real ocean waves and the associated wave spectrum. Nevertheless, in machine learning tasks, a *proxy* to train against is needed. The study by Mittendorf et al. [13] considers wave estimates by the Miros system Wavex[®] as the ground truth. This convention is retained herein, and the results presented in the next section relies directly on the outcome produced by Mittendorf et al. [13].

As already introduced, Wavex has the directional wave spectrum as the final output and examples were shown in Fig. 1. Another example, corresponding to the instant represented by the yellow dot in Fig. 3, is presented in Fig. 7.

The specific directional spectrum is equivalent to a condition represented by the following integral wave parameters: $H_s = 4.0$ m, $T_p = 11.3$ s, $T_z = 8.1$ s, and $D_m = -125^\circ$ (bow oblique waves).

While the WBA also yields a directional wave spectrum as output, it is impractical to compare the actual spectra, say WBA vs. Wavex, and the assessment that follows in Section 5 is, as already mentioned, entirely made for the integral wave parameters H_s , T_p , T_z , and D_m , noticing that the relative mean wave direction follows a slightly different convention than seen from Fig. 7. Thus, $D_m \in [0^\circ - 360^\circ[$ measuring counter-clockwise but still with 180° as head sea.

4.5. Integration limits and discretisation of frequency and direction

As introduced in Section 3.1, the computation of integral wave parameters requires a specified set of cut-off frequencies applied as integration limits. In the analysis presented in the next section, results of the WBA are obtained with lower and upper cut-off frequencies being 0.01 Hz and 0.30 Hz, respectively, and with the frequencies distributed evenly in between, i.e. $\omega_{WBA} = 2\pi \cdot [0.01, 0.02, \dots, 0.30]$ rad/s. It is noted that the particular range of frequencies corresponds exactly to the range of frequencies for which the transfer functions have been computed, cf. Fig. 6. Specifically, as inferred from the figure, it can be seen that the lower cut-off frequency appears reasonable, while the upper cut-off frequency could have been chosen slightly higher, since the sway response, in particular, has not dropped to zero entirely at some of the wave encounter angles. Obviously, nothing can be concluded from this without detailed analyses being beyond the present scope, so here it is left to note that generally any chosen set of cut-off frequencies, potentially, can influence the result of the wave buoy analogy, but in the concrete case study the impact appears relatively small. For the WBA, the wave direction μ , and thus also the wave encounter angle, is spaced evenly on the interval from 0° to 360° , with a 10° spacing.

The directional wave spectra of Wavex come in the same frequency-directional discretisation as the spectra of the WBA. Conversely, the wave parameters in the ERA5 database are obtained from wave spectra of a finer resolution [33], including a higher cut-off frequency, than those of the WBA. In the case of ERA5, however, exact application was made of the downloaded wave *parameters*, directly from the database [25], as the directional spectra were not worked with, which means that the results of ERA5 have not been “manipulated”.

It is a natural fact that most of the energy in developed and developing wave systems at open sea, *and* with a relevance for ships and larger floating structures, generally is carried by waves of a frequency lower than about 0.3 Hz, corresponding to wavelengths of about 17 m, assuming deep-water conditions. This can, for instance, be seen in all of the cases in Fig. 1 (strictly attention should be on Wavex spectra alone as they are not affected by the ship as a low-pass filter). Nonetheless, it is beyond discussion that small discrepancies may be introduced because of the selected set of integration limits and the used discretisation; regardless of “physical limits” (resulting from the low-pass filtering effects of a ship) or “mathematical limits” imposed for practical and consistency reasons. Above all, these considerations should be kept in mind as a contributing factor to the overall uncertainty reflected in the analysis

presented in the following section. For interested reader, it is noteworthy that a smaller sensitivity study about the consequence of lowering the upper cut-off frequency in assessments of ERA5 directional wave spectra has been made by Nielsen and Dietz [18]. Although a different data set was studied,³ it was found that the effect on significant wave height (reflecting the *amount* of energy), by lowering the upper cut-off frequency from 0.55 Hz to 0.20 Hz, was insignificant, with relative deviations between the actual and the filtered values being less than approximately 5% at most and around 1%–2% for the vast majority of cases.

5. Results and discussions

5.1. No conditioning

The “raw” results of the WBA is presented in Fig. 8 that compares the *unconditioned* WBA-estimates with corresponding estimates obtained from Wavex, repeating that the comparison is made for the integral wave parameters alone. It is evident from the comparison that significant scatter appears, although the markers for each of the four parameters $\{H_s, T_p, T_z, D_m\}$ to some degree distribute around the line of identity. A comparison between ERA5 estimates and the unconditioned estimates of the WBA is included in Appendix C for convenience, see Fig. C.19, while a similar comparison between Wavex and ERA5 follows in the next section.

5.2. Validation using wavex output as conditions

As the next step, the proposed methodology can be validated by conditioning directly on the Wavex outcome, cf. Section 3. In this case, Figs. 9–12 show the results; noticing that each integral wave parameter is considered in its own figure. Moreover, results of ERA5 are included to assess the uncertainty associated with the Wavex estimates. As the immediate observation, it is seen that

³ The investigation was made studying sea states as encountered during a seven-days crossing of the North Pacific (Yokohama to Los Angeles) with significant wave heights in the range from 1 m to 5 m.

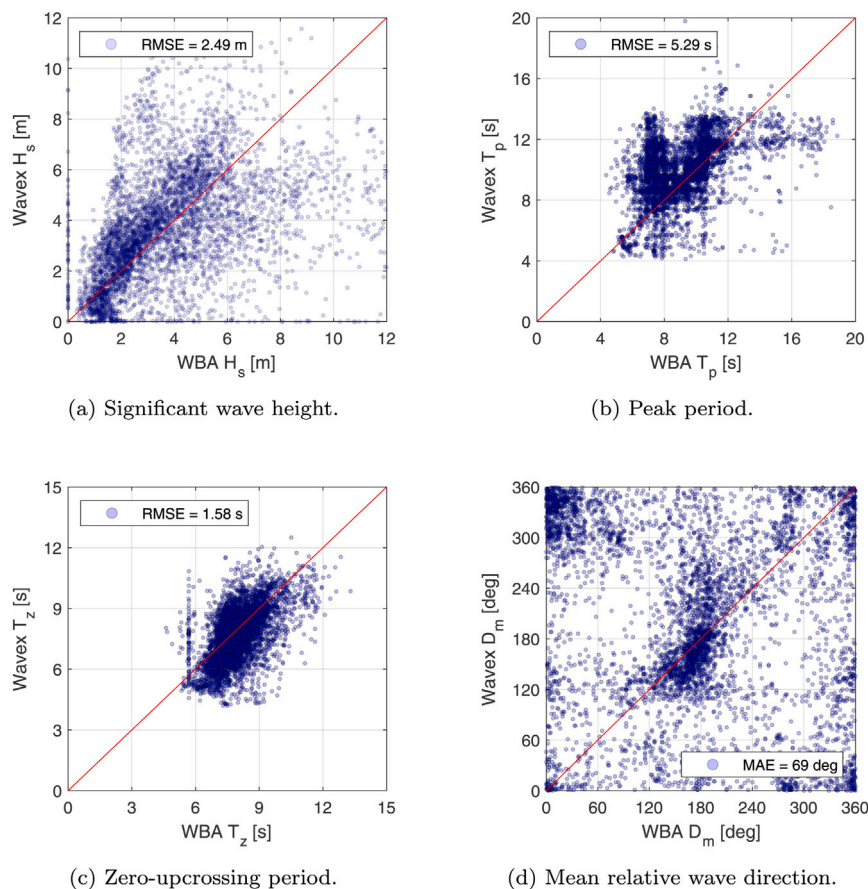


Fig. 8. Comparison of integral wave parameters (H_s , T_p , T_z , and D_m) derived from Wavex and from **unconditioned** wave spectrum estimates by WBA.

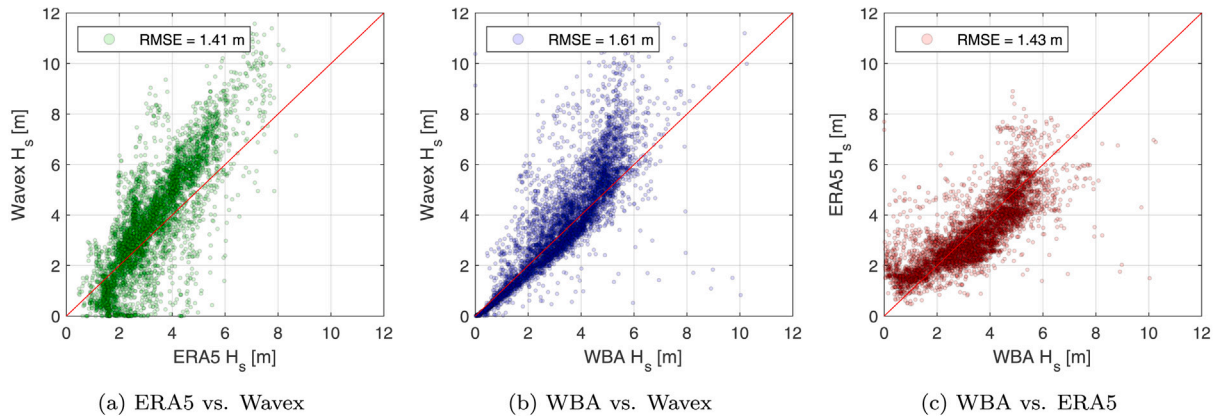


Fig. 9. Significant wave height comparison. Wavex-conditioned estimates by the WBA.

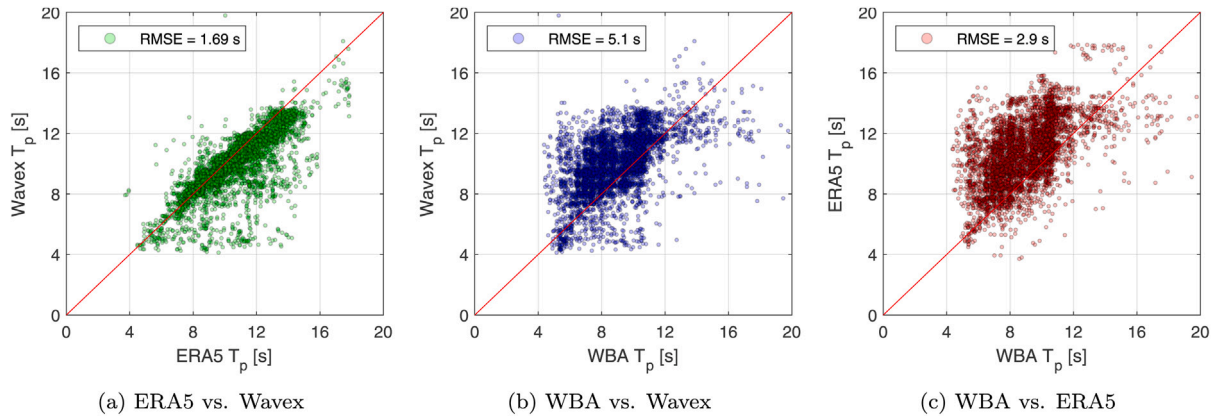


Fig. 10. Peak period comparison. Wavex-conditioned estimates by the WBA.

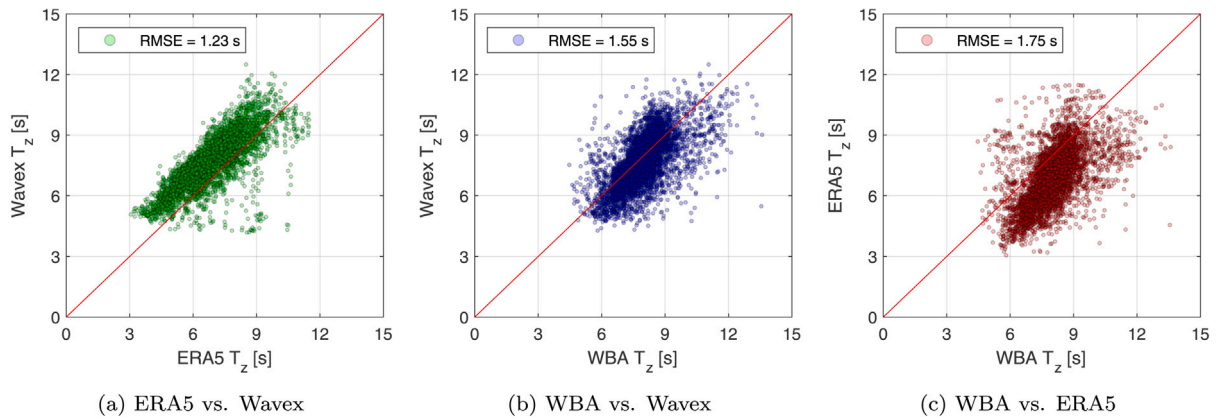


Fig. 11. Zero-upcrossing period comparison. Wavex-conditioned estimates by the WBA.

the agreement between estimates of the WBA and corresponding ones by Wavex, refer to subplots (b) in Figs. 9–12, is improved. This is the case for all wave parameters; however, the improvement is much better/larger for the significant wave height and the mean relative wave direction. This particular finding comes as no surprise since the WBA wave spectrum estimates were conditioned exactly on H_s and D_m , see Eq. (13a) and Eqs. (13b), (13c), respectively.

What also stands out from Fig. 9(a), specifically, is the bias (dependent on H_s) observed in the comparison between Wavex and ERA5; Wavex under-estimates for lower H_s and over-estimates for higher H_s , relative to ERA5. On the other hand, it is interesting

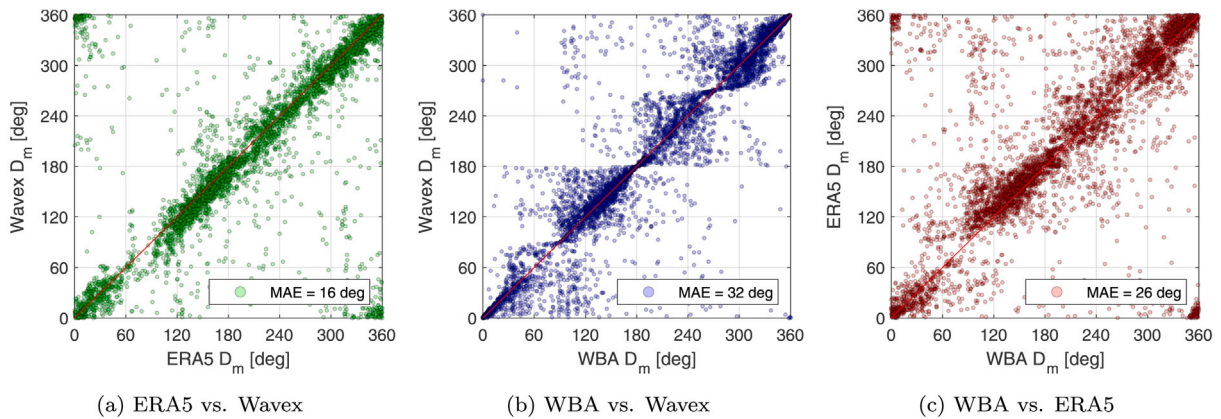


Fig. 12. Mean relative wave direction comparison. **Wavex-conditioned** estimates by the WBA.

to note that a similar bias does not manifest in Fig. 9(c), comparing WBA against ERA5, despite the fact that the WBA-estimates are conditioned on Wavex outcomes. This particular behaviour, inferred as a form of robustness, in the WBA estimates might be explained because of the physics-based approach built into the solution, resulting from the use of transfer functions. However, additional investigations in this direction will be needed to make firm conclusions. At this stage, the central point is that, in the comparison with ERA5, the outcome of the WBA, conditioned on Wavex-results, improves significantly, yet a bias is absent. Here, attention should also be brought to the comparison between the *unconditioned* WBA estimates and the corresponding ERA5 estimates, cf. Appendix C.

As previously introduced, the WBA often yields inconsistent, not to say incorrect, estimates of the mean relative wave direction, refer to subplot (d) in Fig. 8. It is clear, however, that the proposed methodology leads to significant improvements of D_m , reflected by subplots (b) and (c) in Fig. 12. As a practical remark, it has not been thoroughly investigated what leads to the peculiar behaviour in the corresponding estimates of Wavex and WBA (subplot b), reflecting “cuts” roughly at 90°, 180°, and 270°; noticing that similar apparent cuts cannot really be observed in the other comparisons (Wavex vs. ERA5 and ERA5 vs. WBA). In concrete terms, the cuts in Wavex vs. WBA are explained because of a tendency by the WBA to estimate the mean relative wave direction (D_m) over a wider range of directions, specifically where the Wavex estimate is around, respectively, 90°, 180°, and 270°. As indicated, it will take more comprehensive investigations, in a future study, to clarify *why* this happens.

5.3. Benchmarking of machine learning-conditioned estimates

The final assessment concerns the use of ML-based estimates as conditions in Eqs. (13a)–(13c). In the study by Mittendorf et al. [13], the dataset was split into training and validation sets, where the latter was exclusively used for the model assessment; i.e. reflecting the unseen data. The validation set makes up 20% of the total (and filtered) dataset, and comprises $K_v = 956$ samples; reminding that a *sample* in this context consists of a 25-min set of MRU time series and the corresponding Wavex wave spectrum. The training set, on the other hand, has a sample size of $K_t = 3823$. Neural networks are not scale invariant, thus, the individual

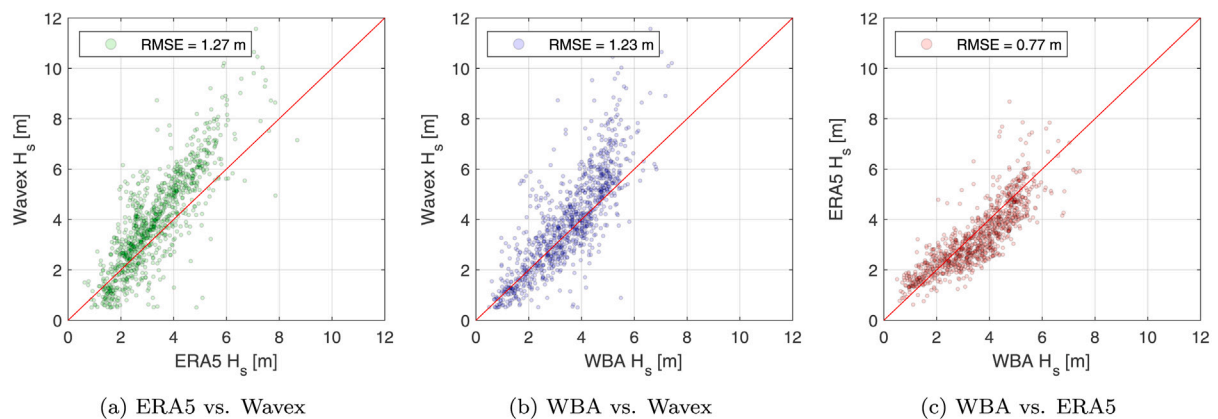


Fig. 13. Significant wave height comparison. **ML-conditioned** estimates by the WBA.

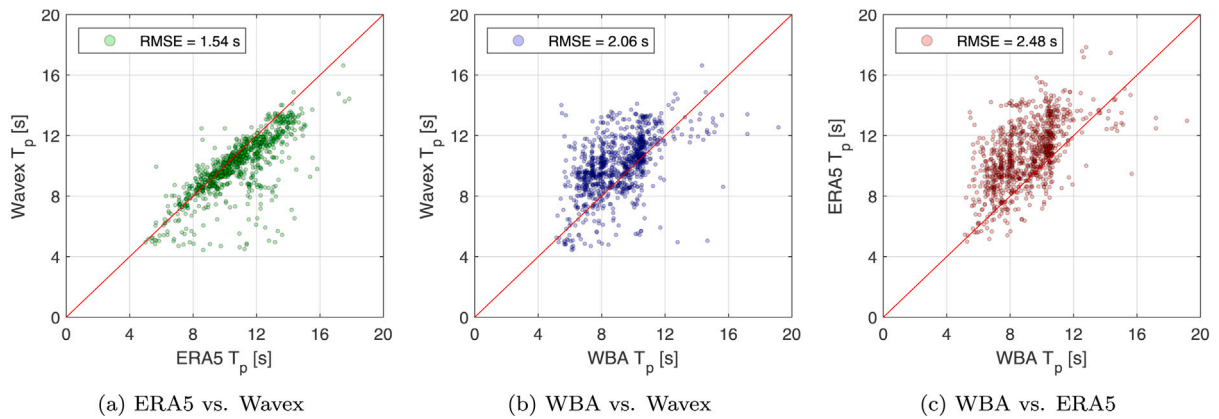


Fig. 14. Peak period comparison. **ML-conditioned** estimates by the WBA.

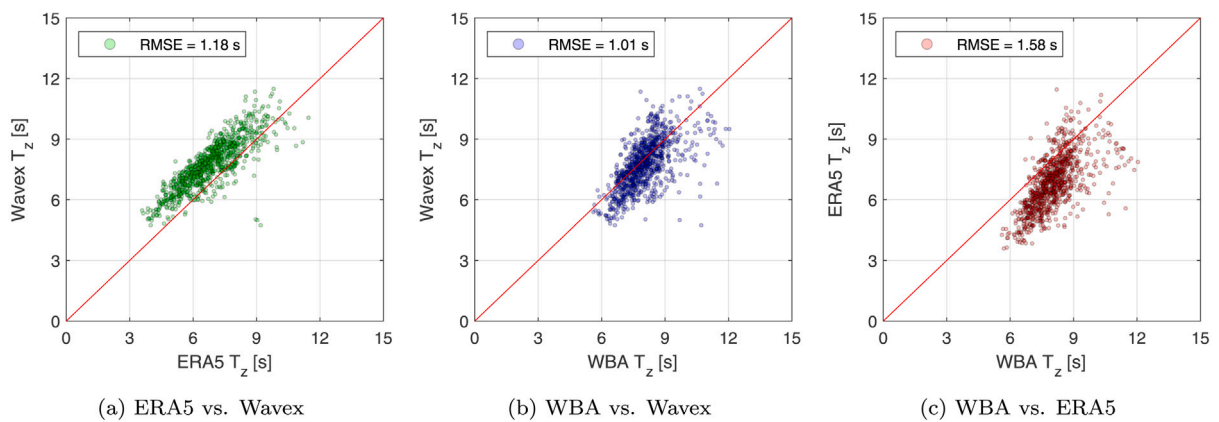


Fig. 15. Zero-upcrossing period comparison. **ML-conditioned** estimates by the WBA.

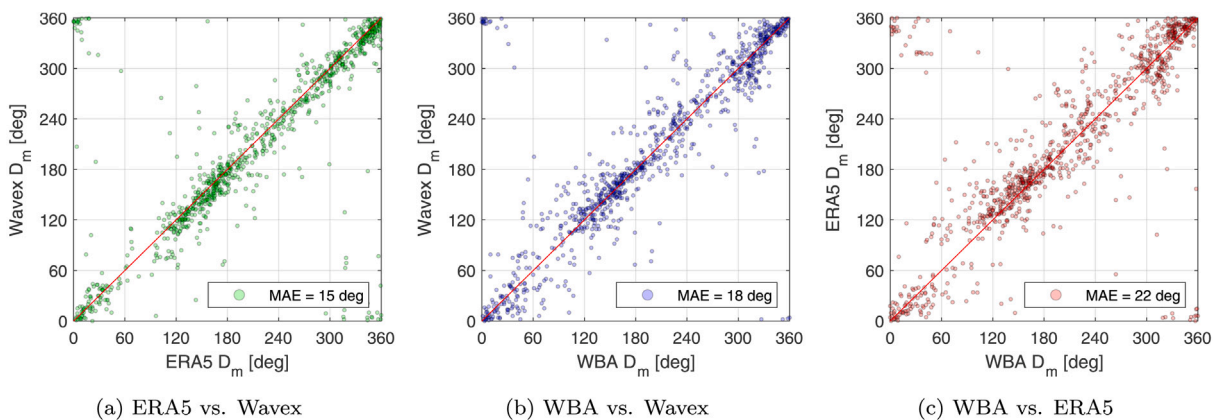


Fig. 16. Mean relative wave direction comparison. **ML-conditioned** estimates by the WBA.

sequences as well as the elements of the output vector were normalised according to the extreme values of the training set to improve performance [13].

Figs. 13–16 present the comparisons of integral wave parameters when the WBA estimates are conditioned on the outcome of the convolutional neural network, considering the validation data exclusively. Overall, it can be noticed that the performance of the proposed method is good. In fact, in this particular situation, the conditioned WBA estimates, focusing on H_s , leads to the lowest value of RMSE in the comparison against ERA5, cf. Fig. 13. Otherwise, the agreements between WBA estimates against Wavex

and ERA5, respectively, appear similar to the observations made in the previous subsection, where a conditioning on the Wavex estimates was made. However, a one-to-one consideration cannot be made due to the reduced dataset considered in the present subsection based on outcome from the machine learning model.

5.4. Prediction of vertical bending moment

It has already been noted that it has little meaning to speak about a *ground truth* when assessing wave estimates in general. Nevertheless, the previous subsections and their comparisons of estimates by Wavex vs. WBA vs. ERA5 indicate good agreement, all inherent uncertainties considered in the single estimates. In terms of application, a very last comparison can be made for assessing the discrepancy, relative to a ground truth, of the actual wave spectrum estimates by Wavex and the ML-conditioned WBA estimates. To do this, the spectrum of the wave-induced vertical bending moment (VBM) has been theoretically computed (predicted) by using the estimated wave spectrum of Wavex and WBA, respectively, and the available transfer function for the VBM, see Eq. (14).

Outcomes of such predictions are presented in Fig. 17 for two arbitrarily selected samples. In the single plot, a direct comparison can be seen between the measured spectrum and the corresponding predicted ones; noticing that the frequency-of-encounter domain is (must be) considered. The legend includes information about the root-mean-squared-normalised-error (rmsne), defined according to

$$\text{rmsne} = \sqrt{\frac{1}{L} \sum_{l=1}^L \frac{(\hat{S}_l - \tilde{S}_l)^2}{\hat{S}_{\max}^2}} \quad (22)$$

where $\hat{S}_{\max}$ is the maximum (i.e. peak) value of the measured spectrum with ordinates indicated by the wide-hat, whereas the predicted spectral ordinates are indicated by the wide-tilde; reminding that L is the number of discrete frequencies considered.

As such, the rmsne yields a measure of the average deviation, compared to the peak value of the measured spectrum, over the spectral ordinates. It is seen that, for the given samples, the prediction with WBA is best for the one case (subplot a), whereas it is opposite for the other case (subplot b). Computing the statistics, based on all samples ($K_v = 956$), the following values are obtained for the average rmsne and the associated standard deviation

$$\text{WBA: mean(rmsne)} = 0.208, \quad \text{std(rmsne)} = 0.147 \quad (23a)$$

$$\text{Wavex: mean(rmsne)} = 0.282, \quad \text{std(rmsne)} = 0.436 \quad (23b)$$

Hence, it can be inferred that the predictions made using the wave spectrum estimates by WBA perform best, having the smallest rmsne, on average, and also the least variation. A graphical assessment, revealing the same overall finding, is shown in Fig. 18, subplots (a) and (b), where the predicted 0th order spectral moment, cf. Eq. (19), is compared to the measured m_0 , all $K_v = 956$ samples considered. Note that the RMSE, included in the legend, is computed by its standard definition, in dimensional form, equivalent to Eq. (20).

It is noteworthy that the above findings with respect to the predicted vertical bending moment are not necessarily unexpected. In fact, even the use of the *unconditioned* wave spectra by the WBA gives better predictions than the use of Wavex spectra does, cf. Fig. 18(c). Specifically, it is noted that, in this case, the RMSE between measured and predicted m_0 has a value of $2.56\text{e-}02$ (GNm)², as compared to $2.77\text{e-}02$ (GNm)² for Wavex. For the more detailed assessment, comparing the average deviation of the spectral ordinates, the statistics of the root-mean-squared-normalised-error (rmsne) are: mean(rmsne) = 0.246 and std(rmsne) = 0.275, which also reflect better outcomes than obtained by Wavex, cf. Eq. (23b). The reason for these findings has to with the particular *ship response model*, relying on the use of measurements and transfer functions from the *same* ship. In this sense, the “real

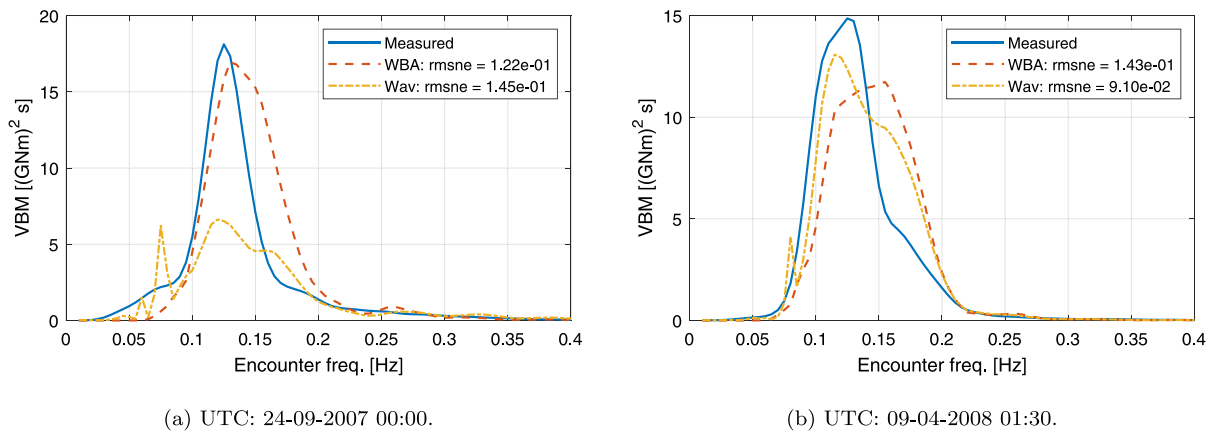


Fig. 17. Direct comparison of the measured spectrum and corresponding predicted ones of the wave-induced vertical bending moment amidship by WBA and Wavex (Wav).

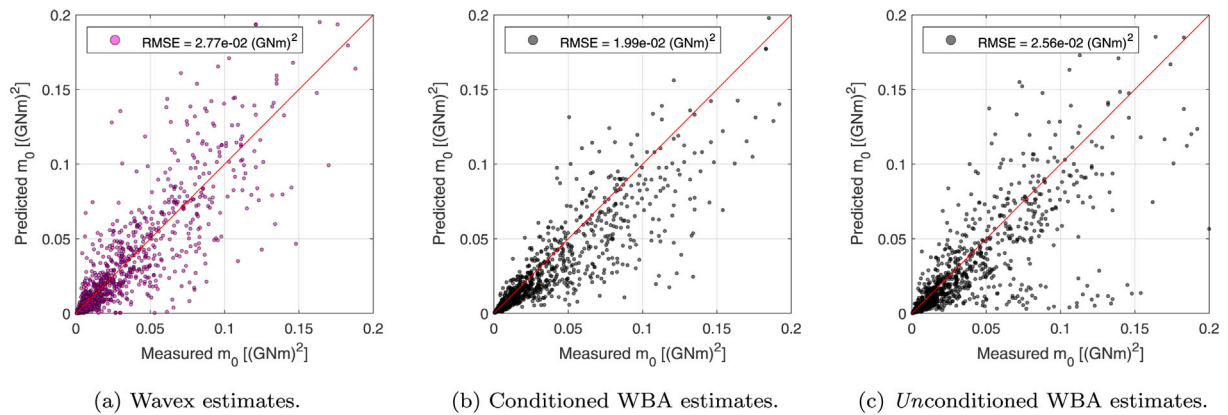


Fig. 18. Comparison of measured and predicted m_0 of the wave-induced vertical bending moment amidship.

test” would be to use the estimated wave spectra for predicting a response *not* being part of the particular ship response model; for instance, one could think of predictions of a wave-induced response of *another* ship. On the other hand, this sort of assessment would be of little practical relevance, leaving that unnoticed. Above all, the central point of the preceding discussion is that the proposed hybrid framework, based on the conditioned wave spectrum estimation, gives good predictions of integral wave parameters as well as responses; and, what is more, the predictions are significantly improved compared to the original (unconditioned) method.

5.5. General remarks and proposed works for the future

It was a central aim of this study to establish a *non-parametric* estimate of the directional wave spectrum. Although numerical studies in the literature very often assume a parameterised wave spectrum, say, a generalised JONSWAP spectrum, possible overlaid with a spreading function to account for short-crested waves, it was evident from examples of estimated directional wave spectra (see Fig. 1) that, in reality, the *real*⁴ wave spectrum may be far off compared to an idealised wave spectrum having just one peak and with the energy density distributed symmetrically around the peak directional-wise. Even by assuming an idealised *bimodal* directional wave spectrum, the results in Fig. 1 suggest that predictions of wave-induced responses, like the vertical bending moment, made with such an idealised spectrum are likely to be inconsistent with predictions made using Wavex or the ML-informed Bayesian version of the WBA developed herein. Clearly, it could be relevant to assess and quantify what the consequences would be if, say, predictions similar to those in Section 5.4 were made actually assuming an idealised and parametrised wave spectrum. In this context, it should be kept in mind that response prediction is feasible only by assuming an idealised wave spectrum if integral parameters alone, say, $\{H_s, T_p, D_m\}$, are available; and this is indeed the case for most (if not all) of the established machine-learning methods available in the literature.

In the developed hybrid framework, it was decided to make use of existing (machine learning) results from a previous study [13] and these results were in turn relying on wave estimates from Miro’s wave radar system Wavex. It is important to stress, however, that the machine learning framework itself, and thus the conditioning of the directional wave spectrum, can use *any* available source of integral wave parameters. From this, it appears interesting to train the convolutional neural network with integral parameters from the ERA5 database, and subsequently apply the outcome as the conditions on the wave spectrum estimate by the WBA. This is a study left for the future.

To follow the discussion in the past paragraph, it is imagined that the developed hybrid framework can run online, i.e. in real-time, and be part of, say, a shipboard decision support system. In this respect, it is essential to keep in mind that the conditioning of the wave spectrum estimate must be made from a *predeveloped* machine learning model, while assimilation in its traditional meaning, using outcome of third-generation spectral wave models (such as ERA5) or satellite altimetry data, is not feasible since such a post-processing analysis can be made *post-voyage* only.

The presented results are somewhat limited in their exploration, and some immediate follow-up studies are planned within the same hybrid framework:

- A dedicated study about the robustness to erroneous results coming out from the ML model. This can be done by specifying a *range* of values around each of the actual wave parameter values produced by the ML-model for any specific sample (time stamp). That is, for a specific sample, say i , with actual output $\{H_s^{ML}, D_m^{ML}\}_i$ from the ML-model, ranges of (“incorrect”) values of H_s and D_m are specified around the actual ones. Subsequently, all of the “incorrect”, unique combinations of H_s and D_m are used for conditioning the WBA estimates to be produced for sample i . Finally, all of the estimated wave spectra, representing the same sample i , can be compared to the estimate, as conditioned on the actual values $\{H_s^{ML}, D_m^{ML}\}_i$, which makes it possible to assess the robustness to erroneous results.

⁴ The ground truth is unknown but for the sake of convenience it is assumed that the resemblance of WBA and Wavex estimates in Fig. 1 is an indication of trustworthy results.

- It is well-known from the literature that the use of certain response combinations can lead to improved wave spectrum estimates depending on operational conditions [e.g. 3,34,35]. The particular dataset, cf. Section 4, makes it possible to study several different response combinations, potentially for automatically deciding on an optimum combination of responses.
- The performance of the parametric method, cf. Section 2, of the physics-based WBA should be investigated. The parametric method relies on an idealised wave spectrum but it is possible to consider a *multi-modal* directional wave spectrum [e.g., 36]. Despite shortcomings of the parametric method, refer to the discussion above, the parametric method has the advantage that the tail of the wave spectrum can be better estimated and this can reduce, to some degree, negative effects of wave filtering, when a ship is used as a wave buoy.
- As reported, there exists no information about the actual draught on the studied voyages, since it was not measured/recorded. It is beyond doubt that the lack of this information adds considerable uncertainty to the produced results, due to draught's general influence on the motion dynamics (added mass, damping, restoring) of a ship. A specialised sensitivity study could therefore address this concern by investigating how much (or little) results are affected by changing the draught to a range of specified values around the transit draught (9.5 m).

6. Conclusions

In this study, a hybrid approach for wave spectrum estimation was presented, fundamentally relying on the wave buoy analogy (WBA). The proposed method combines a machine learning framework with a physics-based solution dependent on available transfer functions. In practical terms, the physics-based solution [4,6] is conditioned on output of integral wave parameters obtained from a deep neural network [13]. The outcome of the hybrid approach is the directional wave spectrum. A comparison of associated integral wave parameters, including significant wave height and mean relative wave direction, against corresponding ones from the ERA5 database [25] showed good agreement. What is more, the WBA outcomes from the new hybrid approach were significantly improved compared to the corresponding outcomes from the original (*unconditioned*) WBA, and the same finding was shown in terms of response prediction using the estimated wave spectra.

While a list of future works is imagined, refer to Section 5.5, one of particular importance will be to assess the robustness of the developed hybrid approach. Here, it is understood that the sensitivity to inaccuracies/inconsistencies in the conditions, obtained via the machine learning model, should be investigated.

CRedit authorship contribution statement

Ulrik D. Nielsen: Conceptualization, Methodology, Software, Formal analysis, Investigation, Validation, Visualization, Writing – original draft & editing. **Malte Mittendorf:** Data curation, Preparation, Methodology, Review, Editing, Writing. **Yanlin Shao:** Data preparation, Review & editing. **Gaute Storhaug:** Data provision, Review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data

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Appendix A. Bayesian (non-parametric) wave spectrum estimation

On the assumption that the ship responses are stationary and linear with the incident waves, the complex-valued frequency response functions $\Phi_i(\omega_e, \beta)$ and $\Phi_j(\omega_e, \beta)$ for the i th and j th responses yield the theoretical relationship between the i th and the j th components of the cross spectra $S_{ij}(\omega_e)$ and the directional wave spectrum $E(\omega_e, \beta)$ through the following integral equation

$$S_{ij}(\omega_e) = \int_{-\pi}^{\pi} \Phi_i(\omega_e, \beta) \overline{\Phi_j(\omega_e, \beta)} E(\omega_e, \beta) d\beta \quad (\text{A.1})$$

where the bar denotes the complex conjugate, and with β being the heading of the ship (relative to the waves) and ω_e being the encounter frequency. It should be realised that the wave spectrum is given in terms of the wave encounter angle, which is justified by letting the wave direction and the course of the ship be given relative to the same datum. Finally, it should be noted that the complex-valued frequency response functions are written as functions of only the heading and the encounter frequency, since the implication of changing operational parameters is understood.

The wave spectrum is advantageously estimated in the wave frequency domain. This means that the speed-of-advance, or triple-valued function, problem needs to be considered. This problem has been properly incorporated by Iseki and Ohtsu [2], for details see [4].

In terms of matrix notation, Eq. (A.1) can be written

$$\mathbf{b} = \mathbf{A}\mathbf{f}(\mathbf{x}) + \mathbf{w} \quad (\text{A.2})$$

The vector function $\mathbf{f}(\mathbf{x})$ expresses the unknown values of the wave spectrum $E(\omega, \beta)$ through a non-negativity constraint $\mathbf{f}(\mathbf{x}) = \exp(\mathbf{x})$, so that $\mathbf{x} = \ln E(\omega, \beta)$. The vector $\mathbf{w}$ is a Gaussian white noise sequence vector with zero mean and variance σ^2 , introduced for stochastic reasons so that the Bayesian modelling is facilitated [22]. The vector $\mathbf{b}$ contains the elements of $S_{ij}(\omega_e)$, and the coefficient matrix $\mathbf{A}$ has elements according to the products of the transfer functions and the derivative of the wave frequency, resulting from the Doppler shift.

In principle, (A.2) can be solved for $\mathbf{x}$ by minimising $\chi^2(\mathbf{x})$ with

$$\chi^2(\mathbf{x}) \equiv \|\mathbf{A}\mathbf{f}(\mathbf{x}) - \mathbf{b}\|^2 \quad (\text{A.3})$$

where $\|\cdot\|$ represents the L_2 norm.

The equation system given by (A.3) is in highly underdetermined and degenerate, which means that the solution is unstable. To overcome this problem, Bayesian modelling is introduced, see [22]. As will be shown in the following, it is possible to evaluate $\mathbf{x}$ by maximisation of the product of the likelihood function and the prior distributions, which must be defined properly. The prior distributions act as a stochastic constraint and are a general character of the model which is known in advance, e.g. [37]. For the model above, the likelihood function is written as

$$l(\mathbf{x}|\sigma^2) = \left(\frac{1}{2\pi\sigma^2}\right)^{\frac{P}{2}} \exp\left(-\frac{1}{2\sigma^2} \|\mathbf{A}\mathbf{f}(\mathbf{x}) - \mathbf{b}\|^2\right) \quad (\text{A.4})$$

where P is the total number of integral equations derived from (A.1) including a number of equations yielding equivalence of energy on the left- and right-hand side of (A.1).

Two prior distributions are taken into account, as originally introduced by Nielsen [6]. Both distributions are a Gaussian smoothness prior distribution which minimises the sum of the second order difference of the unknown vector $\mathbf{x}$ in order to smoothen the change with frequency and direction, respectively, of the wave spectrum. The prior distributions are therefore defined by the minimisation of the functionals

$$\varepsilon_{1mn}^2 = \sum_{n=1}^N \sum_{m=1}^M (x_{m-1,n} - 2x_{m,n} + x_{m+1,n})^2; \quad (x_{0,n} = x_{M,n}, \quad x_{M+1,n} = x_{1,n}) \quad (\text{A.5})$$

$$\varepsilon_{2mn}^2 = \sum_{m=1}^M \sum_{n=2}^{N-1} (x_{m,n-1} - 2x_{m,n} + x_{m,n+1})^2 \quad (\text{A.6})$$

where N and M are the number of discrete wave frequencies and discrete headings, respectively. Thus, considering ε_{1mn}^2 and ε_{2mn}^2 to be normal distributions with zero mean and variance σ^2/u^2 and σ^2/v^2 , respectively, the prior distribution is given in terms of the so-called hyperparameters u and v . In matrix notation the functionals can be written, see e.g. [38],

$$\varepsilon_{1mn}^2 = \mathbf{x}^T \mathbf{H}_1 \mathbf{x} \quad (\text{A.7})$$

$$\varepsilon_{2mn}^2 = \mathbf{x}^T \mathbf{H}_2 \mathbf{x} \quad (\text{A.8})$$

In accordance with [22], the posterior distribution $p(\mathbf{x}|u, v, \sigma^2)$ is proportional to the product of the likelihood function and the prior distribution, which can be written, see [6]

$$p(\mathbf{x}|u, v, \sigma^2) = c \left(\frac{1}{2\pi\sigma^2}\right)^{\frac{P+KM}{2}} |\det(u^2\mathbf{H}_1 + v^2\mathbf{H}_2)|^{1/2} \exp\left(-\frac{1}{2\sigma^2} S(\mathbf{x})\right) \quad (\text{A.9})$$

with

$$S(\mathbf{x}) = \|\mathbf{A}\mathbf{f}(\mathbf{x}) - \mathbf{b}\|^2 + \mathbf{x}^T (u^2\mathbf{H}_1 + v^2\mathbf{H}_2) \mathbf{x} \quad (\text{A.10})$$

and where c is a normalising factor independent of the model parameters $\mathbf{x}$ and the hyperparameters u and v .

The hyperparameters control the trade-off between the good-fit of the solution to the data (i.e. agreement between solution and data) and the smoothness, or stability, of the solution, and the optimum values of the hyperparameters are determined by minimising the control criterion ABIC:

$$\text{ABIC} = -2 \ln \int p(\mathbf{x}|u, v, \sigma^2) d\mathbf{x} \quad (\text{A.11})$$

Appendix B. Inception architecture

In the present paper, the Inception model proposed by Szegedy et al. [39] is applied, which builds on parallel one-dimensional convolutional layers, as these are well-suited for sequential data. Convolutional layers implement spatially shared weights, and [40] present their theoretical intricacies. The development of the Inception architecture underwent several iterations and the version demonstrated by Szegedy et al. [39] is employed in this context. The Inception-v4 has a block-wise structure implementing the so-called Inception module. This module (and architecture) is named after the eponymous science fiction movie. In essence, the Inception module consists of three parts: (1) The bottleneck part is placed in front of the computationally intensive convolutional layers and thus increases efficiency. (2) The second part comprises three parallel convolutional layers with kernel sizes (or receptive fields) of 16, 8, and 4, respectively. Generally, convolutional neural networks benefit from feature extraction at different scales (i.e. with varying filter sizes), which has a resemblance to the working principles of the human visual cortex. (3) The third element acts as an (inner) skip-connection and is made of a pooling and an additional bottleneck layer. Finally, the two individual branches are concatenated, and it is noted that the Inception architecture has an outer residual skip-connection bypassing the Inception module. This concept is adapted from residual networks, which comprise several skip-connections and thus allow for more hidden layers while reducing the risk of vanishing gradients [41]. For detailed information regarding the theory as well as the used architecture reference is given to [13].

Appendix C. Comparison between unconditioned WBA estimates and corresponding ERA5 estimates

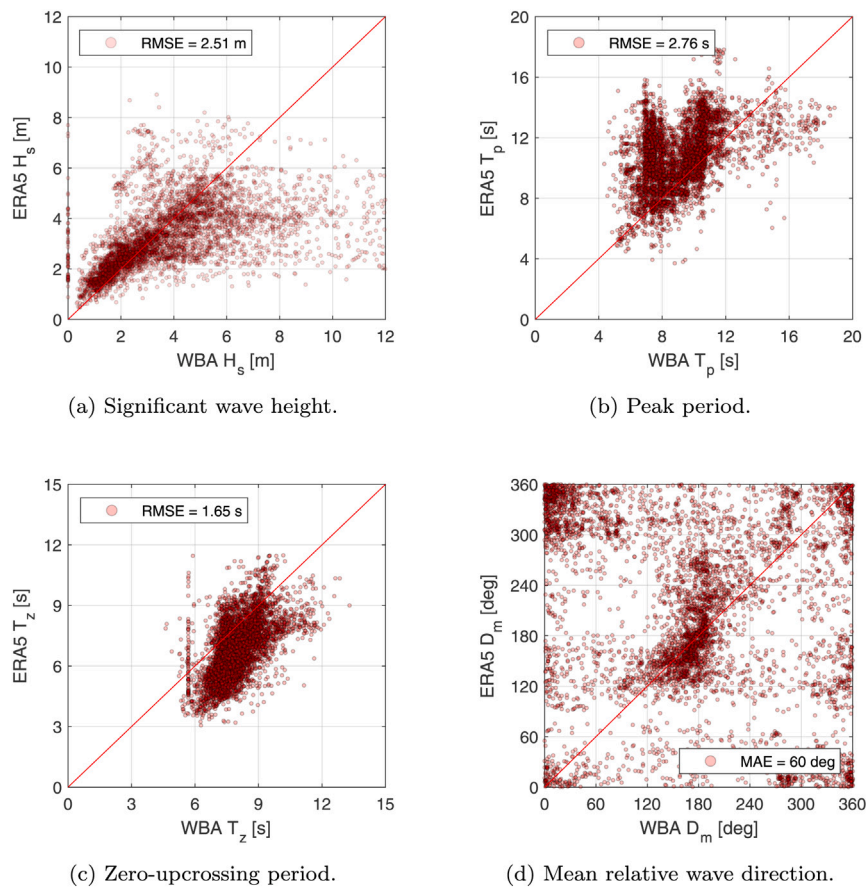


Fig. C.19. Comparison of integral wave parameters (H_s , T_p , T_z , and D_m) obtained from **unconditioned** wave spectrum estimates by WBA and from ERA5.

The comparison between ERA5 estimates and the unconditioned estimates of the WBA is seen in Fig. C.19.

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